COMPGame

COMP3423 · HUMAN–COMPUTER INTERACTION

Student User Manual

How to sign in, work through a topic, and use the AI tutor — a screen-by-screen guide to the whole learning journey.

AUDIENCE	TIME PER TOPIC	YOU NEED	SCREENS SHOWN
Students	About 12 minutes	Your student ID + a password	Captured from the live app

01

What this is

COMPGame flips the usual order. Instead of a lecture and then an exam, you **play with an idea first** in a short game, and **then** find out what stuck. Each of your thirteen topics is one small, self-contained unit that takes about twelve minutes, and everything you do is saved to your account as you go — so you can stop any time and pick up later, even on a different device.

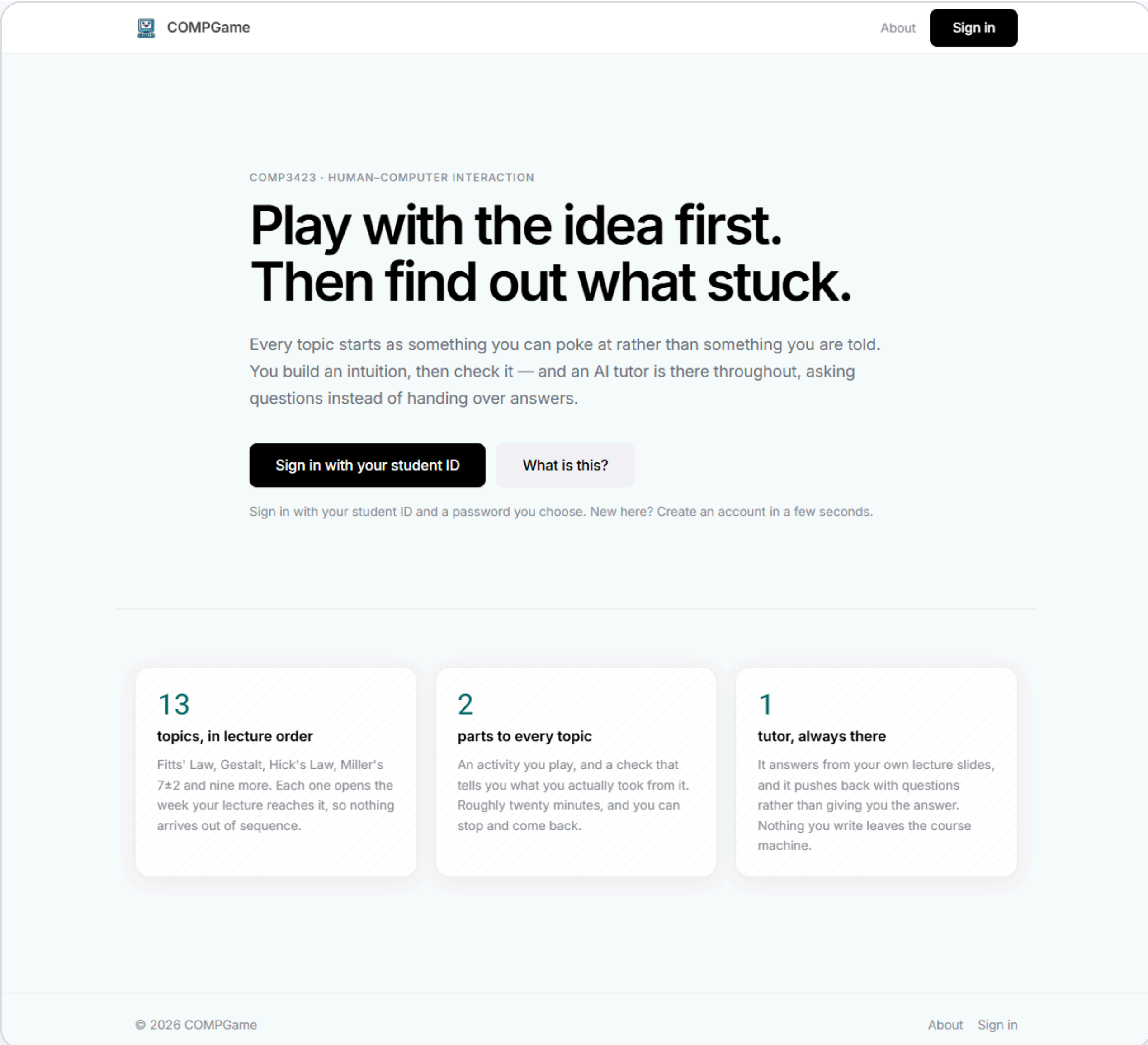


FIGURE 1 The welcome screen. Everything starts here — choose **Sign in** if you already have an account, or **Create account** for your first visit.

> **None of this affects your grade**
> The checks and games are there to help you learn and to help the course see whether learning-by-playing works. There are no marks riding on any of it.

02

Creating your account & signing in

Your login is your **student ID** and a **password you choose**. On your first visit, go to **Create account** and enter your student ID, pick a password, and — if the course has not pre-loaded a class list — choose your section (A, B or C). If the class list is already loaded, your section is filled in for you.

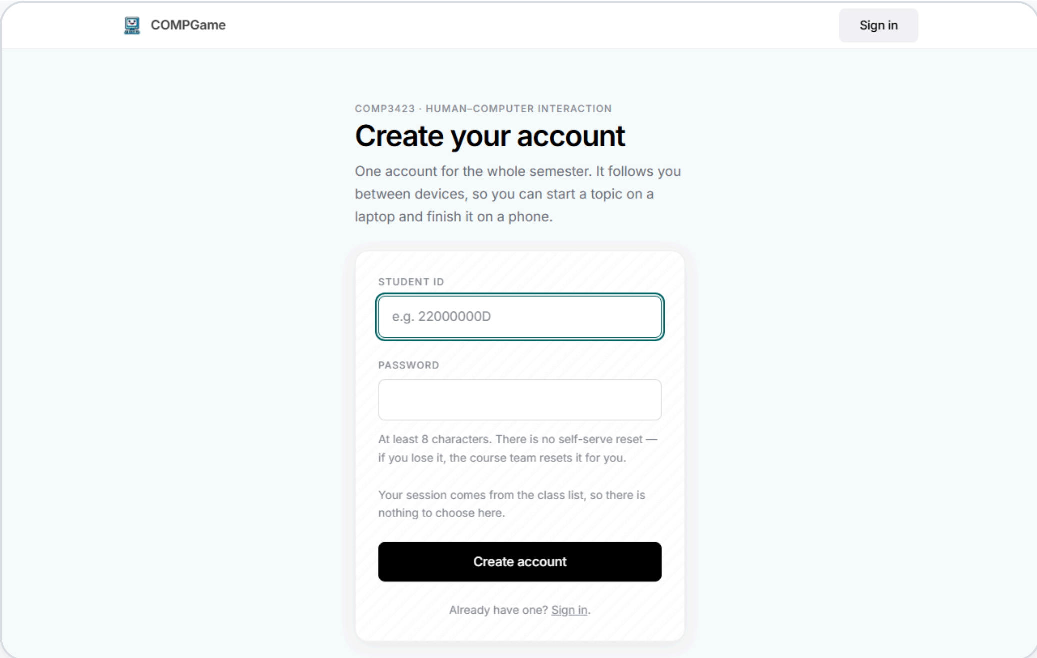


FIGURE 2 Creating an account. Use your real student ID — it is how your progress is kept and, later, how the course links your work together anonymously.

> **Keep your password**
> Passwords are stored one-way and **cannot be looked up** by anyone — not even the course team. If you forget it, a teacher can reset it for you from the course-team panel; nobody can tell you your old one.

After that first time, you just **Sign in** with the same student ID and password. If you ever get a single "that didn't work" message, it is deliberately vague for privacy — check the ID and password and try again.

03

Agreeing to take part

The first screen after your first sign-in is a short consent page. It explains, in plain language, what is recorded and why. Read it, tick the box, and choose **Agree** to continue. Taking part is voluntary — you can **withdraw at any time** from *Account & my data*, which removes your account and ends every session.

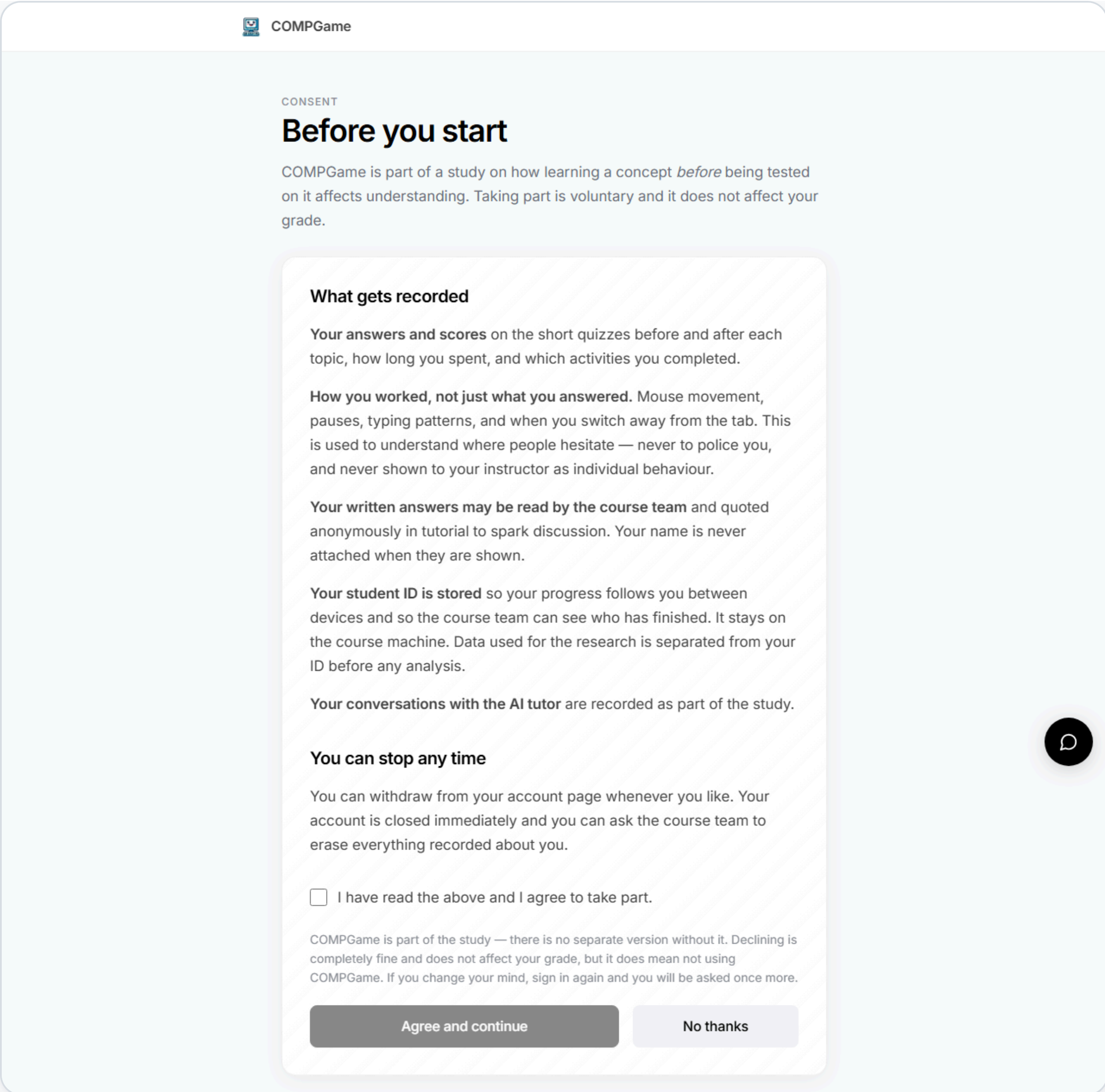


FIGURE 3 The consent screen. Nothing is recorded against you until you agree here.

04

Setting up — avatar, name, and a warm-up

A quick three-step setup gives your account a face and a name, and asks a short warm-up quiz. The warm-up is **not graded** and you are not expected to know the answers — it simply gives the course a starting point to measure against.

SET-UP 1

Pick an avatar

Your character across the whole journey.

SET-UP 2

Choose a name

What the app calls you.

SET-UP 3

Warm-up quiz

A few questions. Answer as best you can.

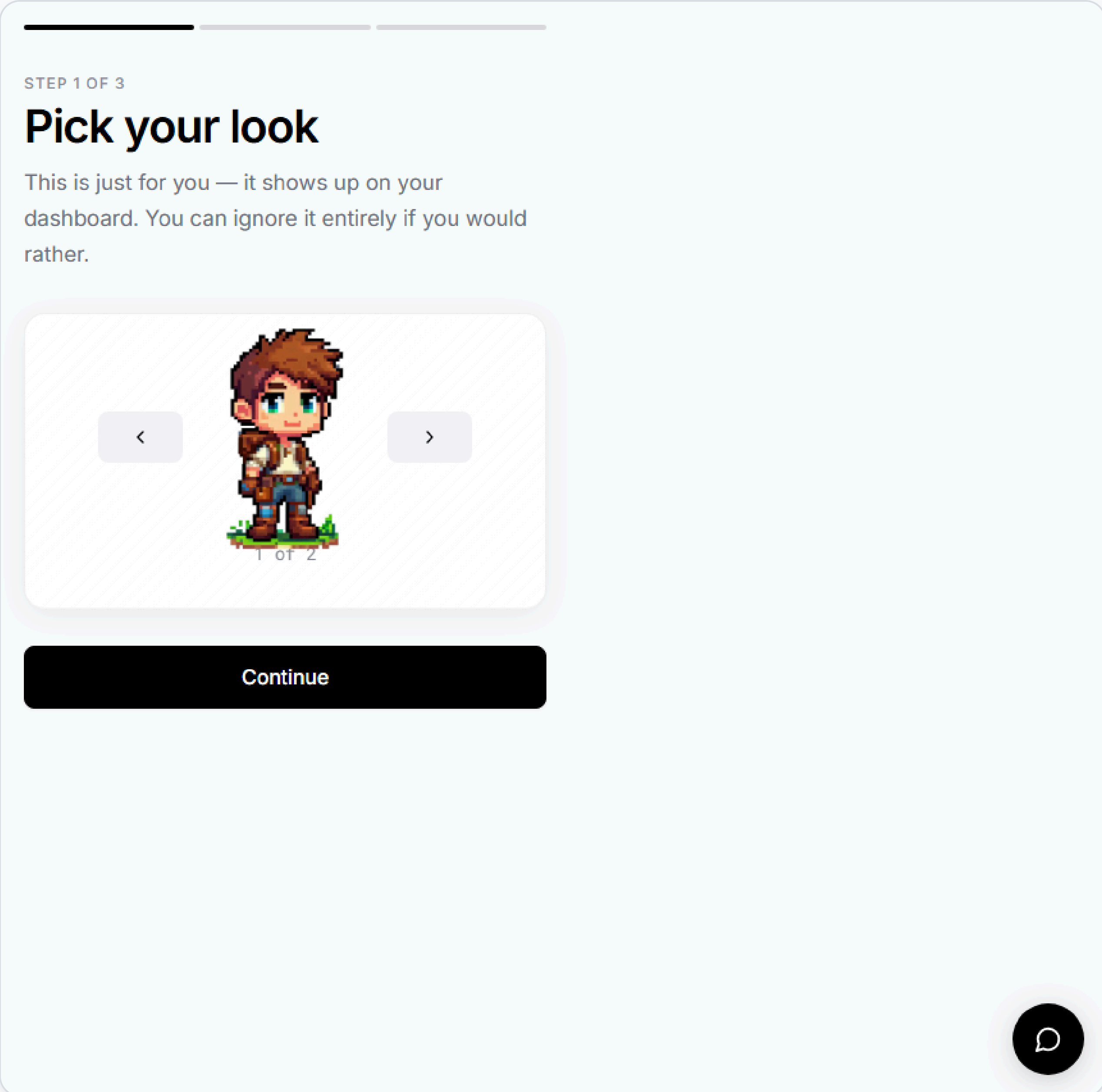


FIGURE 4 Choosing an avatar.

STEP 3 OF 3

Before you start, five quick questions

These tell us what people already know before using COMPGame at all. You are not expected to know them, they do not affect your grade, and you will not be asked them again.

QUESTION 1 OF 5

Fitts' Law predicts that pointing time...

A increases as targets get farther away or smaller

B depends mainly on screen resolution

C is constant regardless of target position

D decreases as the number of targets increases

QUESTION 2 OF 5

Miller's Law states that working memory holds approximately...

A 3 items at a time

B 7 ± 2 chunks

C 15 items if chunked well

D unlimited items with practice

QUESTION 3 OF 5

In Norman's Action Cycle, the 'Gulf of Execution' means...

A the computer takes a long time to respond

B the user cannot tell whether their action worked

C the user cannot figure out how to do what they want

D the interface has too many steps

QUESTION 4 OF 5

The Gestalt principle of Proximity says objects are grouped because...

A they look similar to each other

B they are physically near each other

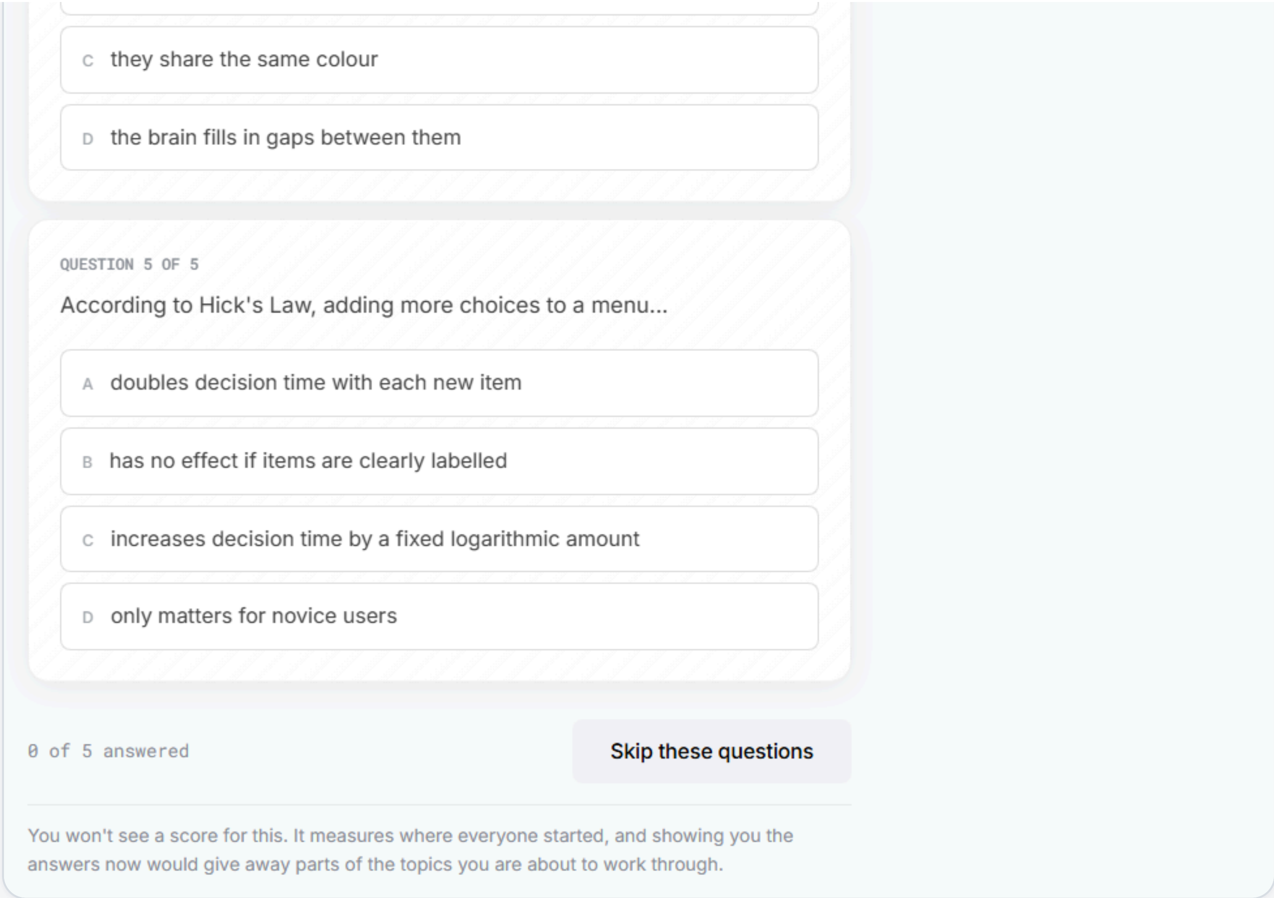


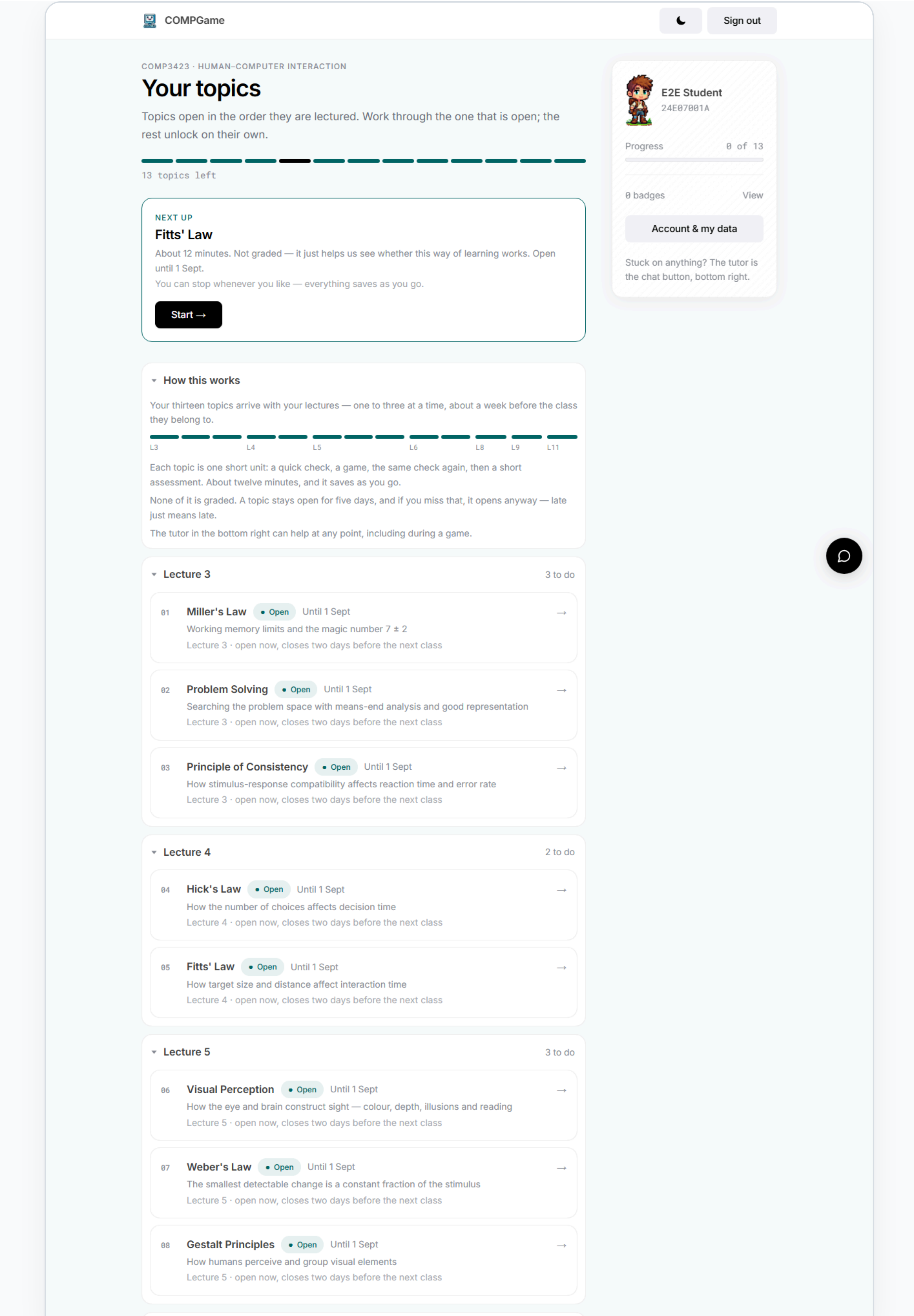
FIGURE 5 The warm-up quiz. Just pick the option that seems right — a guess is fine.

05

Your topics — the home screen

After setup you land on **Your topics**. This is your home base. Topics arrive in the order they are lectured, roughly a week before the class they belong to, so at any moment **one topic is open** and the rest are waiting.

- **Next up** — the card that matters right now. It tells you the topic, roughly how long it takes, and when it closes. Press **Start** to begin.
- **The progress strip** — the row of segments shows all thirteen topics and how far you have come.
- **How this works** — a short explainer you can open any time.
- **The sidebar** — your avatar, badges so far, and *Account & my data* (export or withdraw).


COMPGame
Sign out
COMP3423 · HUMAN-COMPUTER INTERACTION
Your topics
Topics open in the order they are lectured. Work through the one that is open; the rest unlock on their own.
13 topics left
NEXT UP
Fitts' Law
About 12 minutes. Not graded — it just helps us see whether this way of learning works. Open until 1 Sept.
You can stop whenever you like — everything saves as you go.
Start →
How this works
Your thirteen topics arrive with your lectures — one to three at a time, about a week before the class they belong to.
L3
L4
L5
L6
L8
L9
L11
Each topic is one short unit: a quick check, a game, the same check again, then a short assessment. About twelve minutes, and it saves as you go.
None of it is graded. A topic stays open for five days, and if you miss that, it opens anyway — late just means late.
The tutor in the bottom right can help at any point, including during a game.
Lecture 3
3 to do
01 Miller's Law • Open Until 1 Sept
Working memory limits and the magic number 7 ± 2
Lecture 3 · open now, closes two days before the next class
02 Problem Solving • Open Until 1 Sept
Searching the problem space with means-end analysis and good representation
Lecture 3 · open now, closes two days before the next class
03 Principle of Consistency • Open Until 1 Sept
How stimulus-response compatibility affects reaction time and error rate
Lecture 3 · open now, closes two days before the next class
Lecture 4
2 to do
04 Hick's Law • Open Until 1 Sept
How the number of choices affects decision time
Lecture 4 · open now, closes two days before the next class
05 Fitts' Law • Open Until 1 Sept
How target size and distance affect interaction time
Lecture 4 · open now, closes two days before the next class
Lecture 5
3 to do
06 Visual Perception • Open Until 1 Sept
How the eye and brain construct sight — colour, depth, illusions and reading
Lecture 5 · open now, closes two days before the next class
07 Weber's Law • Open Until 1 Sept
The smallest detectable change is a constant fraction of the stimulus
Lecture 5 · open now, closes two days before the next class
08 Gestalt Principles • Open Until 1 Sept
How humans perceive and group visual elements
Lecture 5 · open now, closes two days before the next class
E2E Student
24E07001A
Progress
0 of 13
0 badges
View
Account & my data
Stuck on anything? The tutor is the chat button, bottom right.

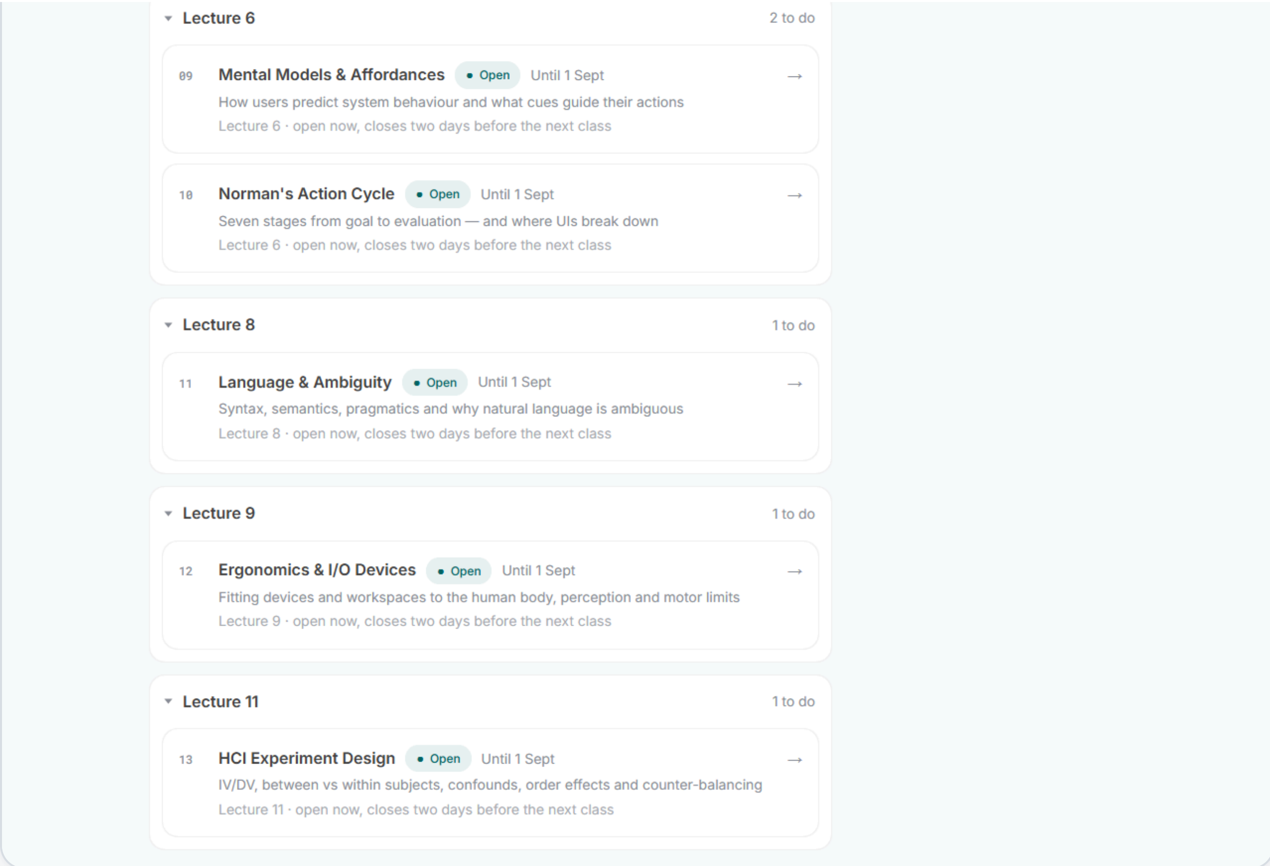


FIGURE 6 The topics home screen. Work through whatever is open; the rest unlock on their own as the term goes on.

> **The tutor is always one tap away**
> The dark chat button in the bottom-right corner opens the AI tutor from any screen — including in the middle of a game.

06

Inside a topic — the learning path

Opening a topic shows a short brief and then walks you through a fixed path. A small strip at the top always shows which step you are on (for example, *Step 3 of 7*), so you are never lost. Each step is recorded the moment you finish it, and **Continue** appears once it has — you do not have to declare yourself done.

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5	STEP 6
Brief	**First check**	**Activity**	**Second check**	**Short answer**	**Tutor**
What the topic is about.	A few questions, before you learn.	Play the game.	The same idea, new questions.	Explain it in your words.	Talk it through.

STEP 7

Done

Badge + replay.

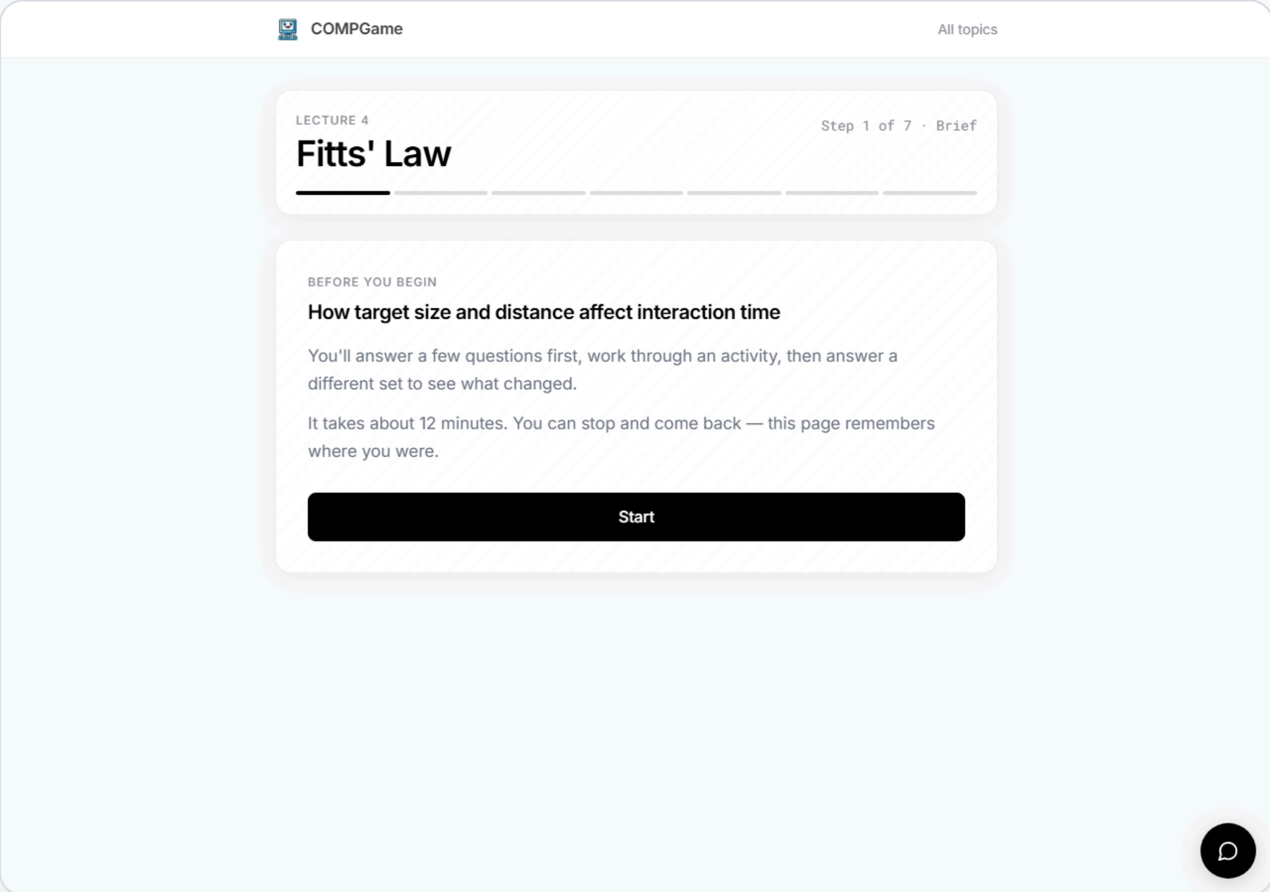


FIGURE 7 A topic brief. The order of the game and the checks can vary slightly between students — that is by design, and you always just follow the path in front of you.

07

The two checks

Every topic has the same short quiz **twice**: once before the activity and once after. You get **one attempt** at each, but you can change your mind freely until you press **Submit**.

- **First check** — reveals nothing. No score, no right answers. That is on purpose, so the second check is a fair measure of what you learned.
- **Second check** — reveals everything: your score, and the correct answer marked on every question.

COMPGame

All topics

LECTURE 4

Fitts' Law

Step 2 of 7 · First check

FIRST CHECK

Before you start

Answer as best you can — you're not expected to know these yet, and this doesn't affect your grade. You'll get one attempt, and no answers until the end of the topic.

QUESTION 1 OF 6

Fitts' Law predicts how long a pointing movement takes. Which two properties of that movement does it use?

- A How far the target is and how wide it is
- B The speed of the hand and the size of the screen
- C How many targets are on screen and how bright they are
- D The user's age and the input device

QUESTION 2 OF 6

In `ID = log₂(A/W + 0.5)`, what does **W** stand for?

- A The number of windows open
- B The width of the target along the direction of approach
- C The weight given to the movement
- D The waiting time before the movement starts

QUESTION 3 OF 6

Two buttons sit the same distance from the pointer. One is 30 px wide, the other 120 px. Which has the higher index of difficulty?

- A They are equal, because the distance is the same
- B The 120 px button, because it covers more area
- C The 30 px button, because a smaller W makes the ratio A/W larger
- D Neither — ID depends only on distance

QUESTION 4 OF 6

A paper reports a target's index of difficulty as **3.2**. Three point two *what*?

- A Milliseconds
- B Pixels
- C Centimetres
- D Bits

QUESTION 5 OF 6

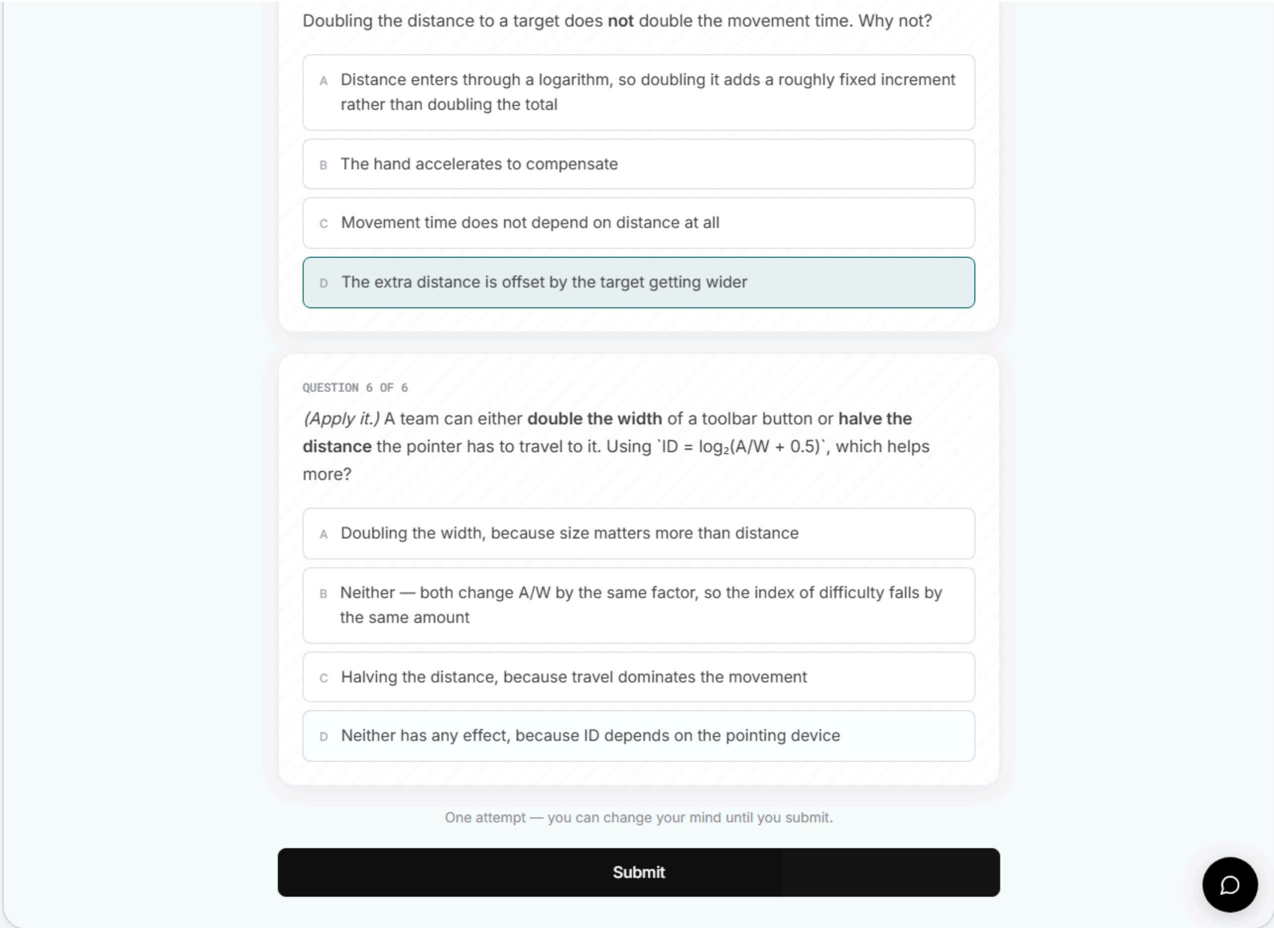


FIGURE 8 The first check. Select one option per question, then **Submit**. You will not see how you did until the very end of the topic.

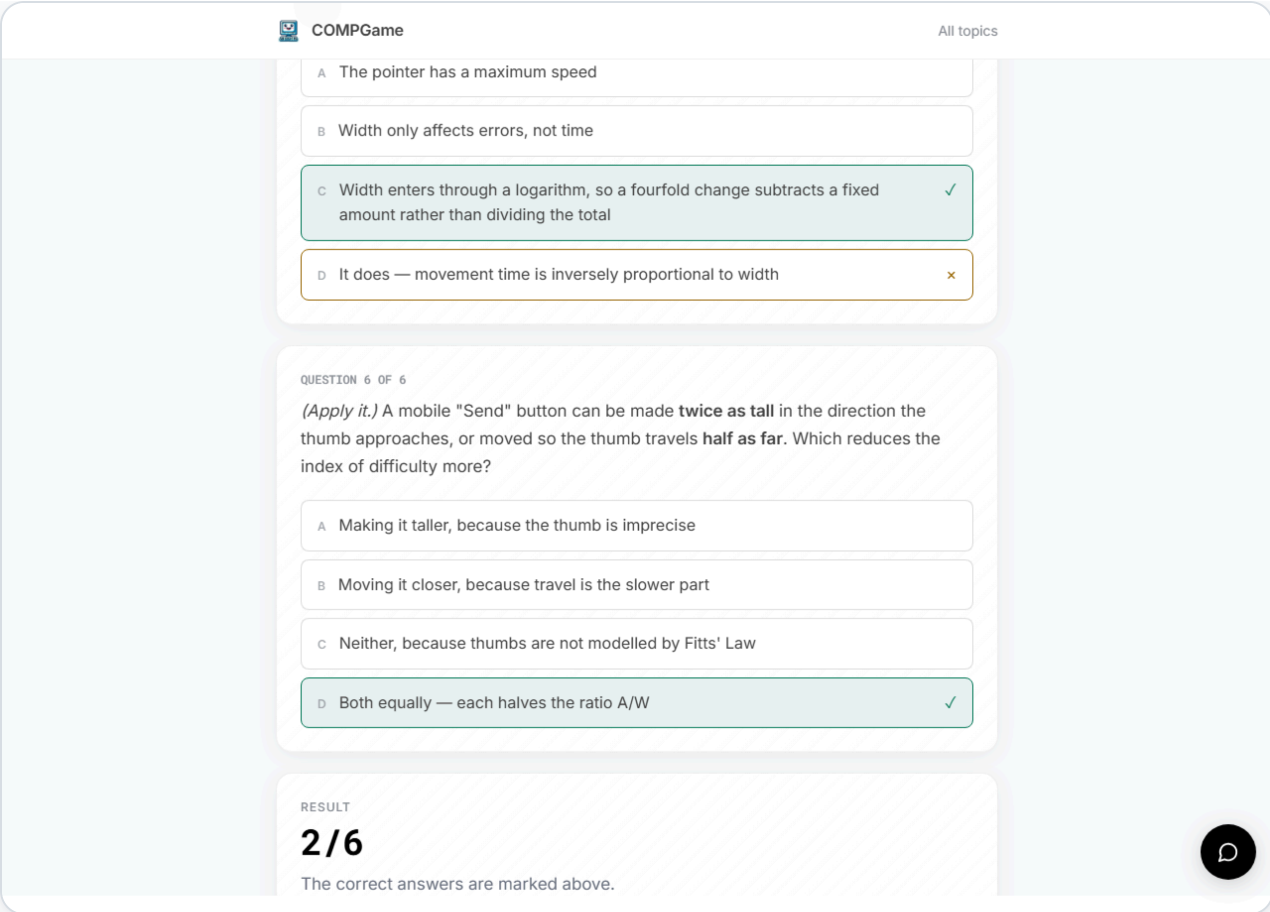


FIGURE 9 The second check, after submitting. A green tick marks the correct answer; a cross marks an option you picked that was wrong. Your score is at the bottom.

08

The activity — playing the game

The heart of each topic is a small game that lets you feel the idea rather than read about it. Read the short intro on the game's start screen, then play. For the best experience, use the **fullscreen** button in the game's bottom-right corner.

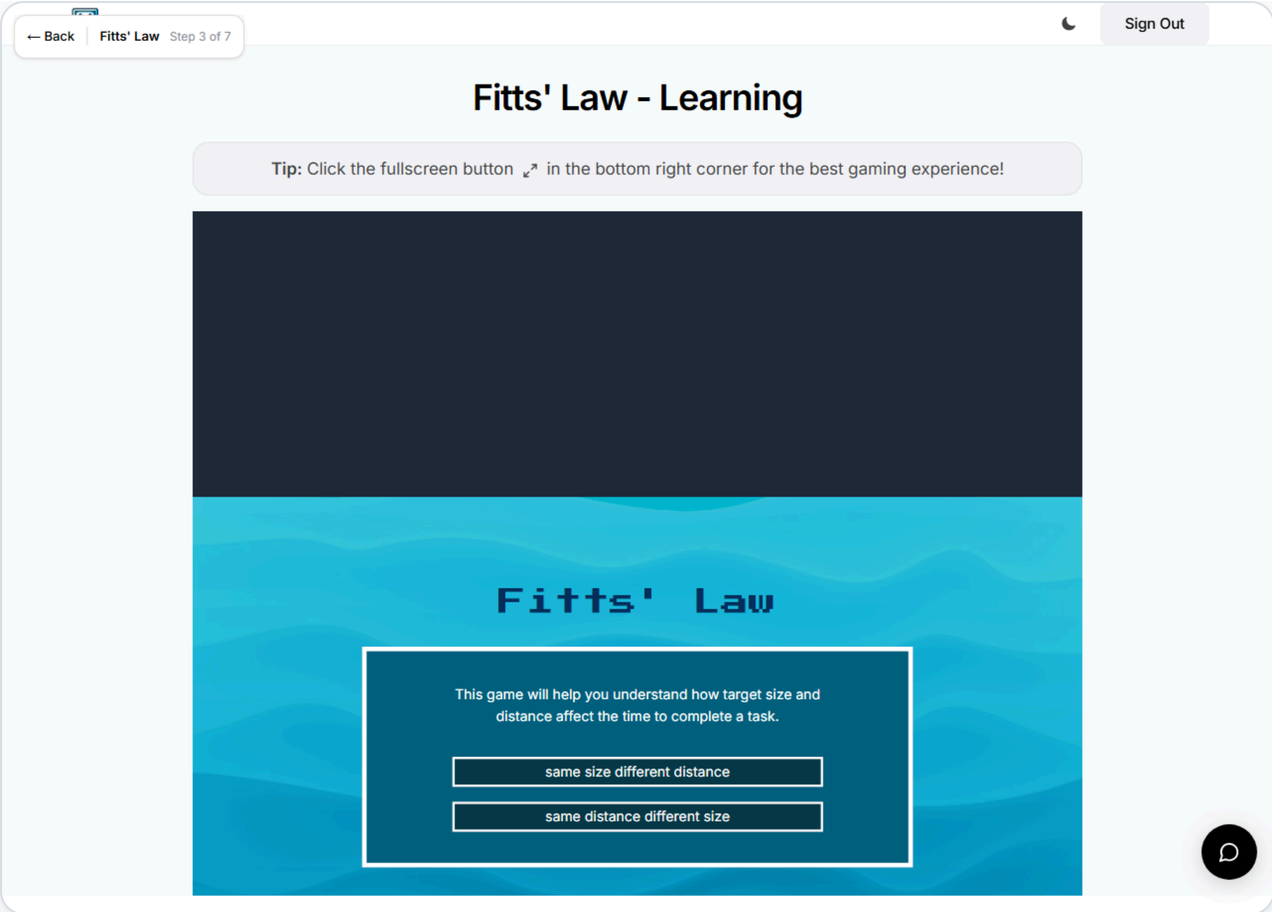


FIGURE 10 A game start screen. Pick a mode and play — the game shows you the concept in action.

> **If a game will not load or record**
> Games are the one place a browser can occasionally misbehave. If one genuinely will not record after you have tried it, the unit offers a small *continue anyway* link so you are never stuck — using it is logged, and it will not cost you anything. When the game records normally, a full **Continue** appears instead.

09

The AI tutor

The tutor is woven through the whole journey, not bolted on. Open it from the chat button any time — including during a game or a check — and ask it about the current topic. It teaches the **Socratic** way, nudging you toward the answer with questions rather than just handing it over; if you would rather see a worked example, **ask it for one** and it will give you it.

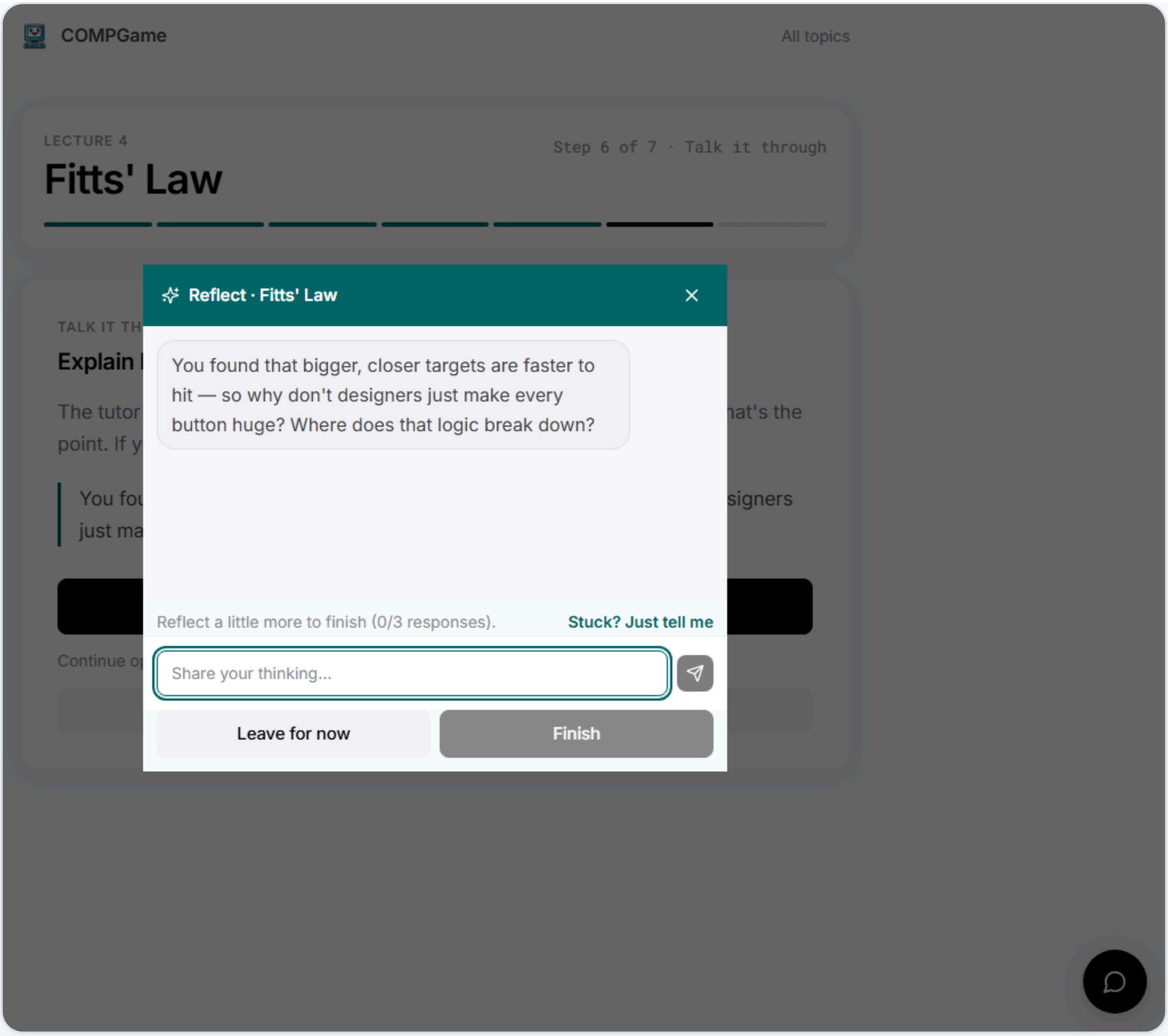


FIGURE 11 The tutor in conversation. It answers from the actual course material, so its help lines up with your lectures.

10

Finishing a topic

When you reach the end you get a short close screen: how your two checks compared, the **badge** you earned, and the chance to **replay both games** as free play. Then it is back to your topics, where the next one is waiting.

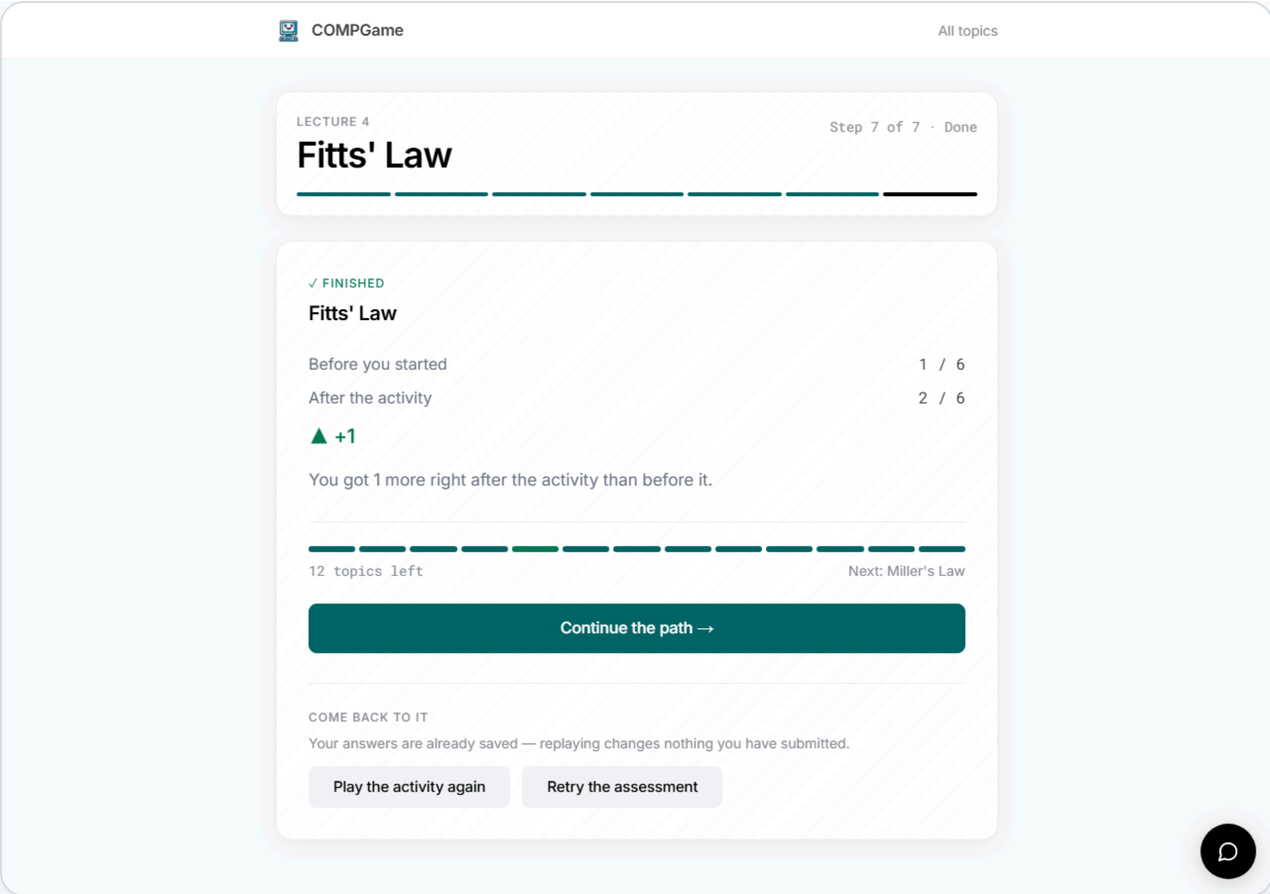


FIGURE 12 The close screen. Your badge records what you did, and the games stay open to play again with no pressure.

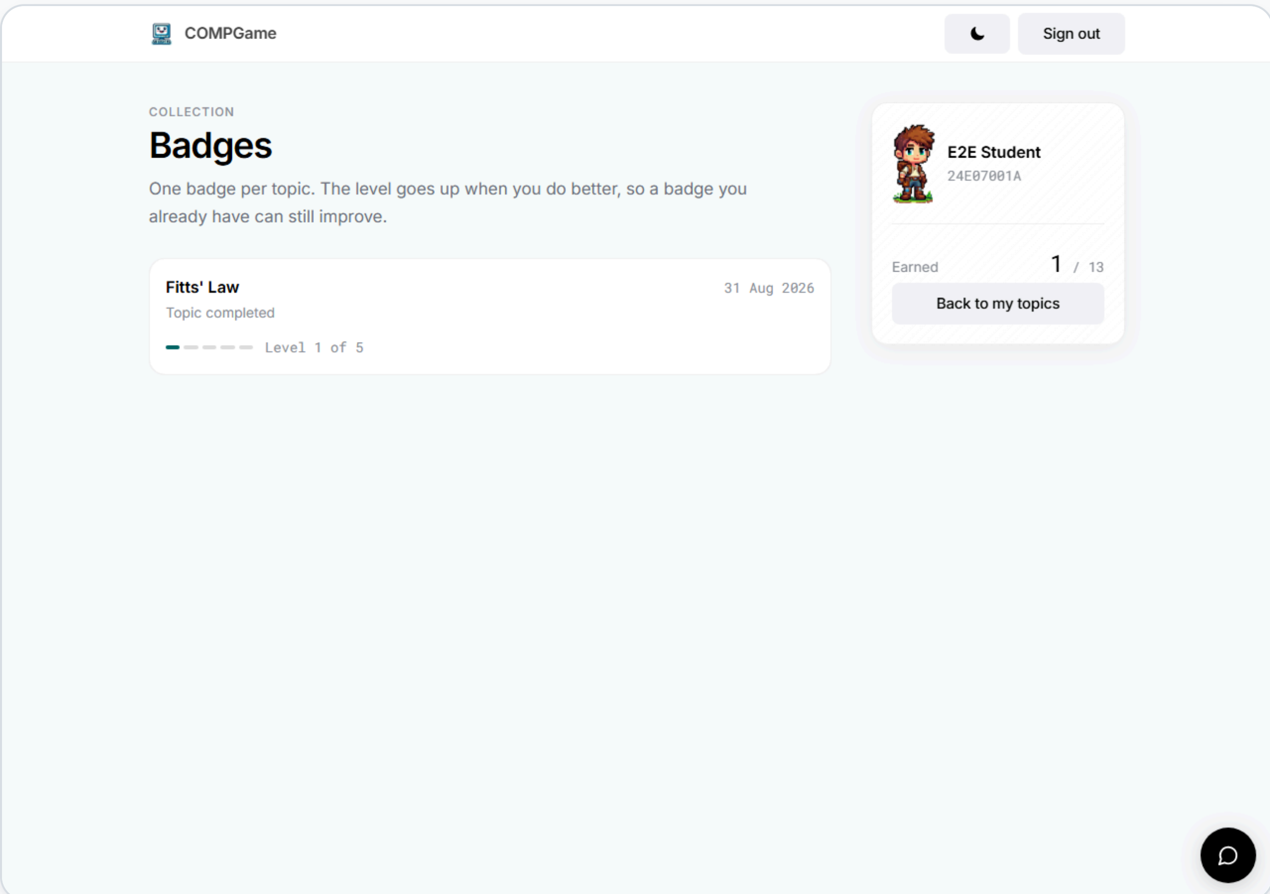
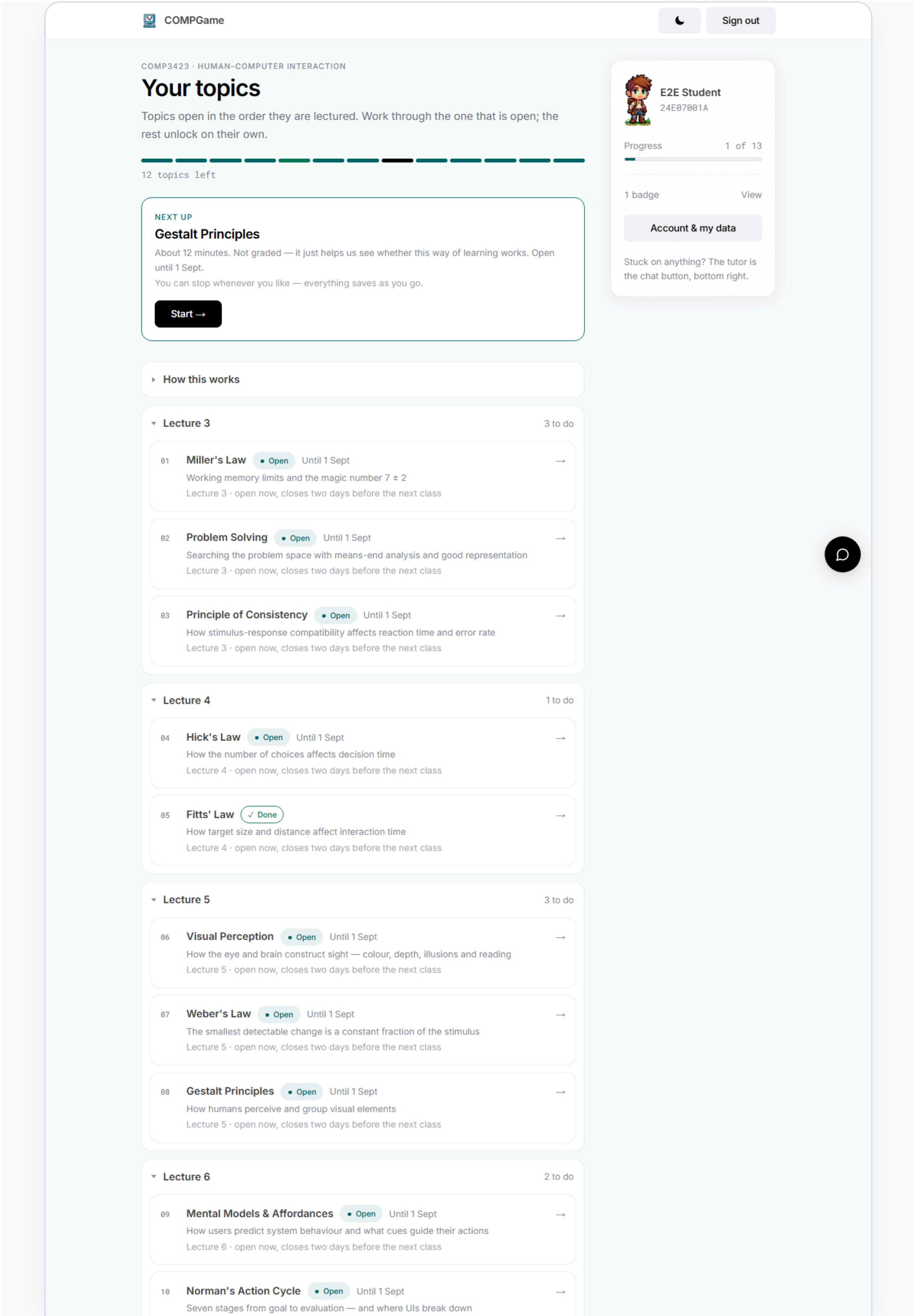


FIGURE 13 Your badges collect on their own page, one story per topic.



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Principle of Consistency
• Open
Until 1 Sept
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Lecture 4
1 to do
04
Hick's Law
• Open
Until 1 Sept
How the number of choices affects decision time
Lecture 4 · open now, closes two days before the next class
05
Fitts' Law
✓ Done
How target size and distance affect interaction time
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3 to do
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Until 1 Sept
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• Open
Until 1 Sept
How humans perceive and group visual elements
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Lecture 6
2 to do
09
Mental Models & Affordances
• Open
Until 1 Sept
How users predict system behaviour and what cues guide their actions
Lecture 6 · open now, closes two days before the next class
10
Norman's Action Cycle
• Open
Until 1 Sept
Seven stages from goal to evaluation — and where UIs break down
E2E Student
24E07001A
Progress
1 of 13
1 badge
View
Account & my data
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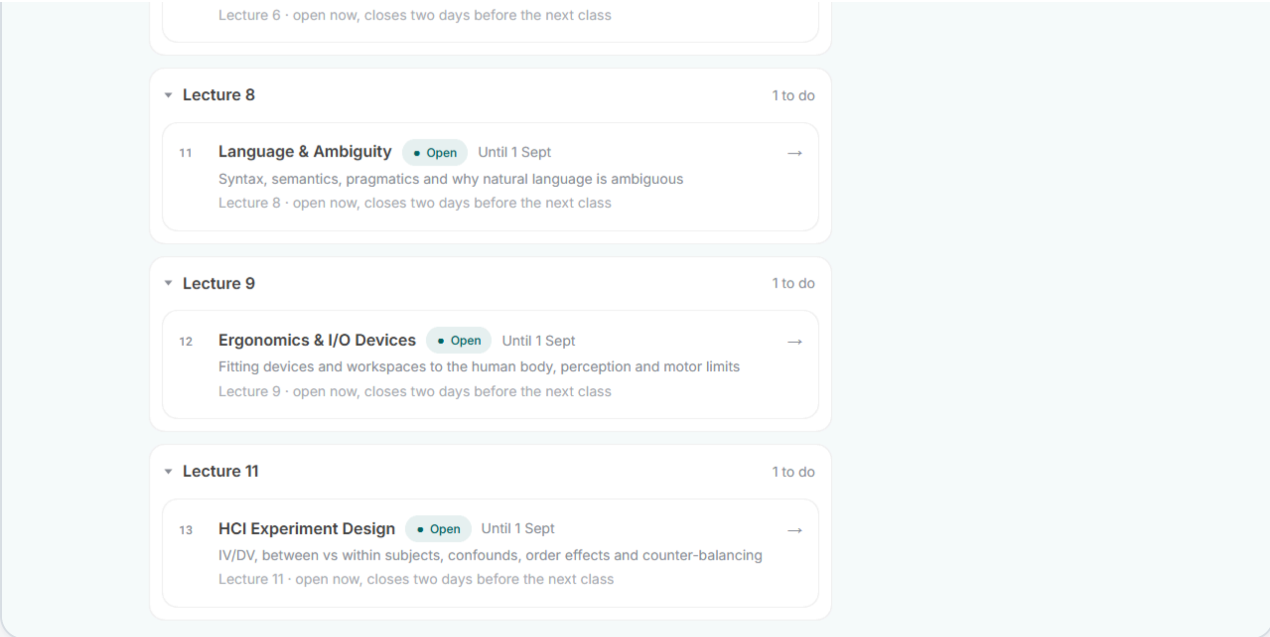

FIGURE 14 Back home, your progress has moved and the next topic is queued up.

11

Good to know

- **It saves as you go.** Close the tab whenever — your place is kept on your account, not just this browser.
- **Any device.** Sign in on a laptop today and a phone tomorrow; your progress follows you.
- **Signed out after 30 minutes idle.** If you walk away, you will be asked to sign in again. While you are actually using the app — even typing a long answer — you stay signed in.
- **A topic stays open for about five days.** Miss it and it opens anyway; late just means late, and nothing is graded.
- **Your data is yours.** Export a copy or withdraw entirely from *Account & my data* at any time.

> **Stuck on anything?**
>
> The tutor (chat button, bottom-right) is the fastest help for the material itself. For anything about your account or access, ask your course team.

COMPGame · COMP3423 Human–Computer Interaction · Student User Manual · Screens captured from the live application.